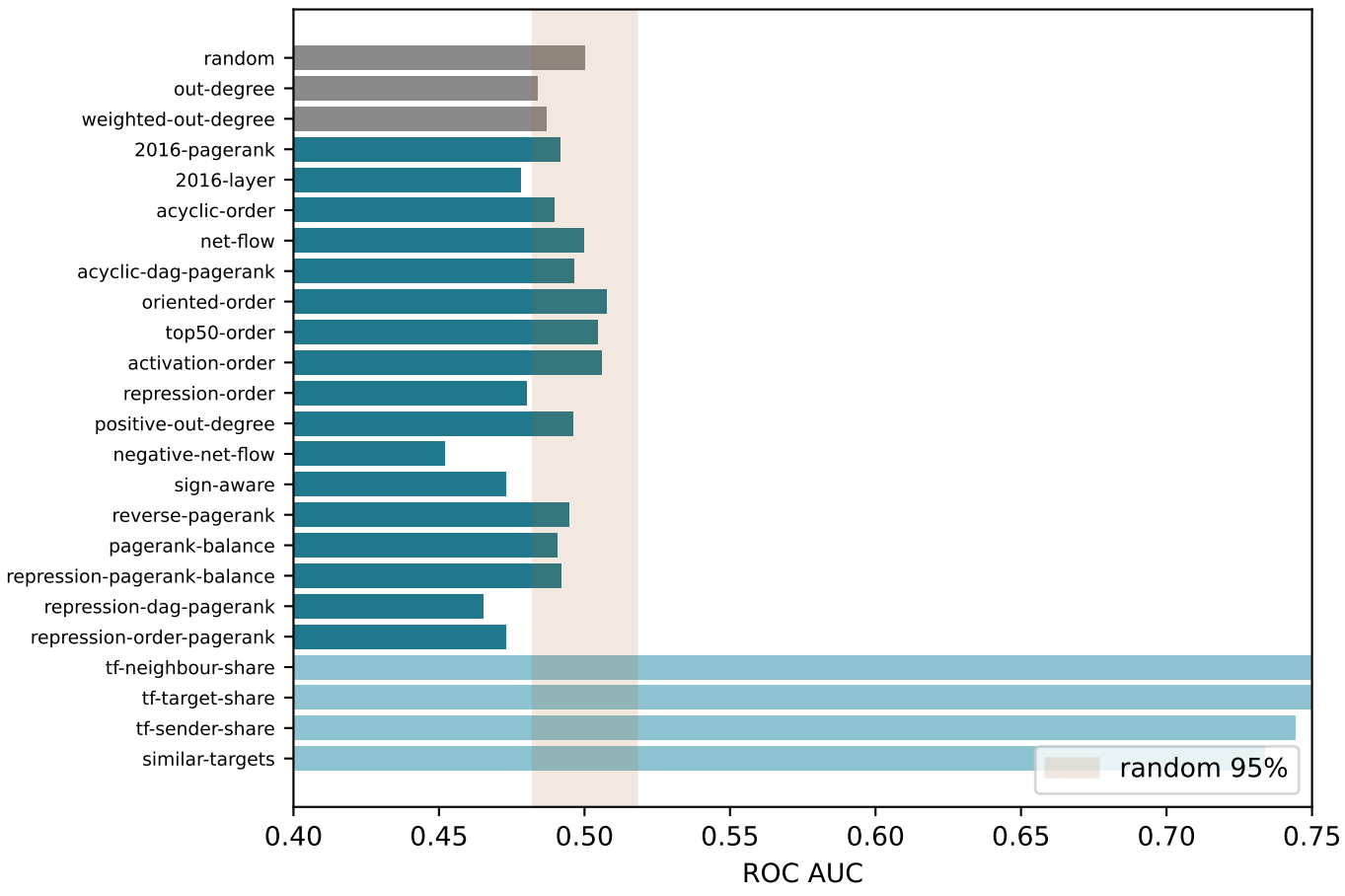
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,969 genes ranked, 1,153 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–16)	0.500 (0.482–0.518)				
out-degree	14	0.484	0.097	0.959	-0.003 (-0.004, -0.002)	< 0.001
weighted-out-degree	16	0.487	0.031	0.928	+0.000 (+0.000, +0.000)	
2016-pagerank	6	0.492	0.920	0.824	+0.005 (-0.010, +0.020)	0.552
2016-layer	9	0.478	0.637	0.992	-0.009 (-0.026, +0.008)	0.312
acyclic-order	13	0.490	0.166	0.875	+0.003 (-0.004, +0.010)	0.478
net-flow	13	0.500	0.166	0.502	+0.013 (+0.000, +0.025)	0.048
acyclic-dag-pagerank	15	0.496	0.055	0.646	+0.010 (+0.003, +0.016)	0.006
oriented-order	9	0.508	0.637	0.210	+0.021 (+0.007, +0.036)	0.004
top50-order	5	0.505	0.968	0.315	+0.018 (-0.000, +0.035)	0.064
activation-order	16	0.506	0.031	0.271	+0.019 (+0.012, +0.027)	< 0.001
repression-order	6	0.480	0.920	0.986	-0.007 (-0.020, +0.007)	0.348
positive-out-degree †	16	0.496	0.031	0.655	+0.009 (+0.007, +0.012)	< 0.001
negative-net-flow †	9	0.452	0.637	1.000	-0.035 (-0.047, -0.022)	< 0.001
sign-aware †	12	0.473	0.254	0.999	-0.014 (-0.020, -0.008)	< 0.001
reverse-pagerank †	11	0.495	0.360	0.708	+0.008 (+0.004, +0.012)	< 0.001
pagerank-balance †	10	0.491	0.494	0.849	+0.004 (-0.009, +0.017)	0.572
repression-pagerank-balance †	12	0.492	0.254	0.806	+0.005 (-0.008, +0.018)	0.416
repression-dag-pagerank †	10	0.465	0.494	1.000	-0.022 (-0.037, -0.008)	0.004
repression-order-pagerank †	11	0.473	0.360	0.999	-0.014 (-0.027, -0.002)	0.022
tf-neighbour-share ‡	100	0.768	< 0.001	< 0.001	+0.281 (+0.255, +0.304)	< 0.001
tf-target-share ‡	96	0.768	< 0.001	< 0.001	+0.281 (+0.256, +0.305)	< 0.001
tf-sender-share ‡	99	0.744	< 0.001	< 0.001	+0.258 (+0.232, +0.282)	< 0.001
similar-targets ‡	97	0.734	< 0.001	< 0.001	+0.247 (+0.222, +0.272)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,846	0.045 (3.18×)	3.62×	1.47×	0.80 (0.79)
resolve_reciprocals → drop_sinks → prune_triangles	30,744	0.005 (1.72×)	3.28×	1.50×	0.36 (0.34)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	25,076	0.004 (1.51×)	2.59×	1.43×	0.32 (0.28)
absolute	31,846	0.045 (3.18×)			
absolute → minimum_feedback_order	30,911	0.034 (2.92×)			
orient_reciprocals ratio=2.0 → absolute	30,592	0.021 (2.24×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,592	0.021 (2.25×)			
top_outgoing k=50 → absolute	31,846	0.016 (4.77×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,847	0.011 (4.33×)			
keep_sign sign=1 → absolute	31,846	0.041 (3.62×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,887	0.029 (3.29×)			
keep_sign sign=-1 → absolute	27,922	0.005 (1.49×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	27,370	0.005 (1.55×)			
keep_sign sign=-1	27,922	0.005 (1.49×)		1.49×	0.23 (0.23)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 1,992,167 edges (435,016 negative), largest strong component 11,968.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 1,992,167 edges (435,016 negative), largest strong component 11,968.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 458,320 edges (403,612 negative), largest strong component 11,525.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 458,320 edges (403,612 negative), largest strong component 11,525; DAG keeps 256,650 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,992,167 edges (0 negative), largest strong component 11,968; DAG keeps 1,565,956 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 1,992,167 edges (0 negative), largest strong component 11,968.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 1,992,167 edges (0 negative), largest strong component 11,968; DAG keeps 1,565,956 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,238,346 edges (0 negative), largest strong component 10,307; DAG keeps 1,229,256 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 489,792 edges (0 negative), largest strong component 11,956; DAG keeps 365,931 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,557,151 edges (0 negative), largest strong component 11,950; DAG keeps 1,171,351 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 435,016 edges (0 negative), largest strong component 9,798; DAG keeps 400,707 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 1,992,167 edges (435,016 negative), largest strong component 11,968.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 1,992,167 edges (435,016 negative), largest strong component 11,968.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 1,992,167 edges (435,016 negative), largest strong component 11,968.

reverse-pagerank: Highest reverse PageRank (raw graph)